Evaluating Place-Based Policy with Staggered Difference-in-Differences

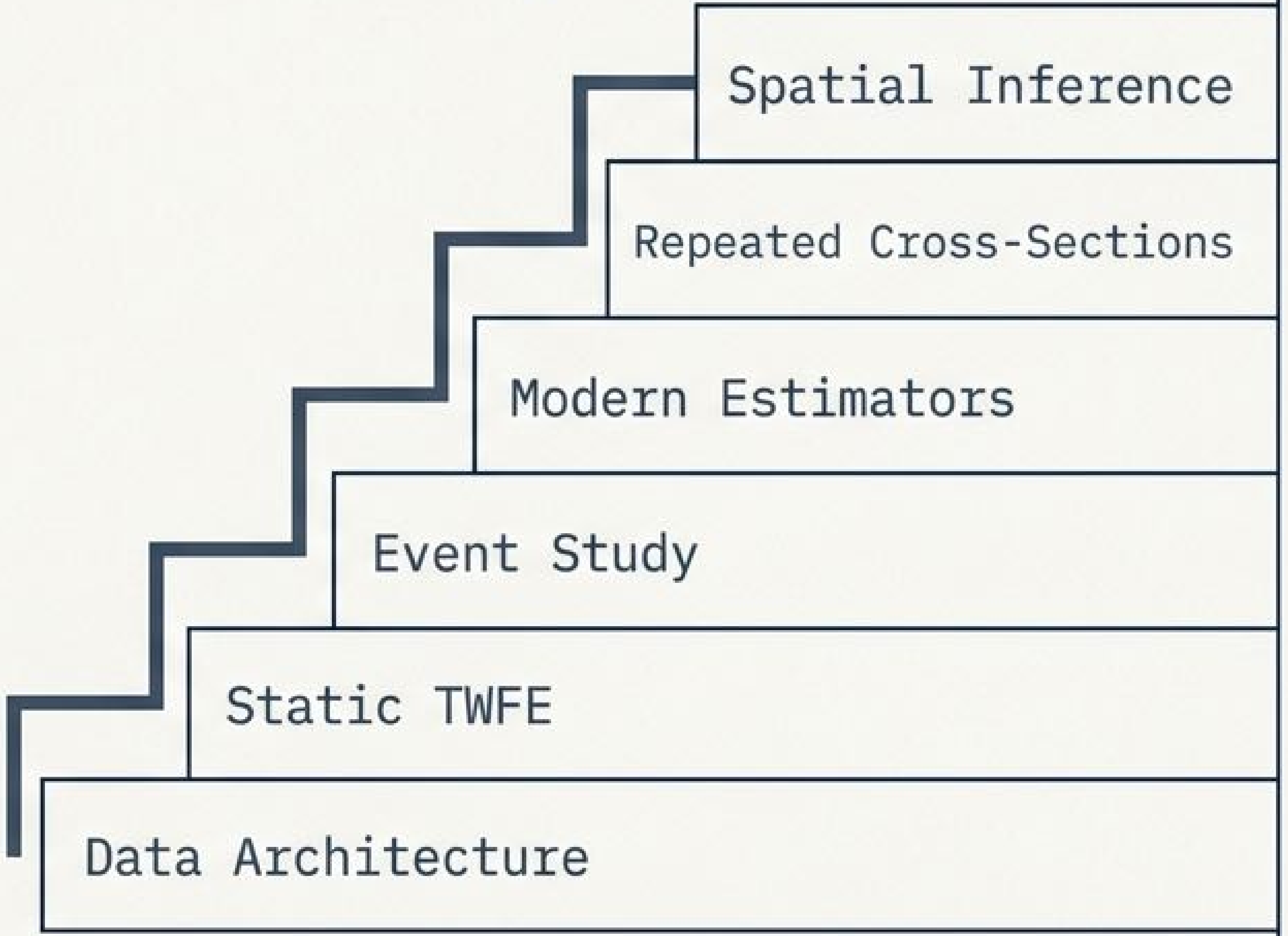
Abstract Summary

A methodological study guide on evaluating the local economic and welfare impacts of Ethiopia's industrial parks using synthetic calibrated data (replicating Huang, Wang & Xu, 2026).

Methodological Scope

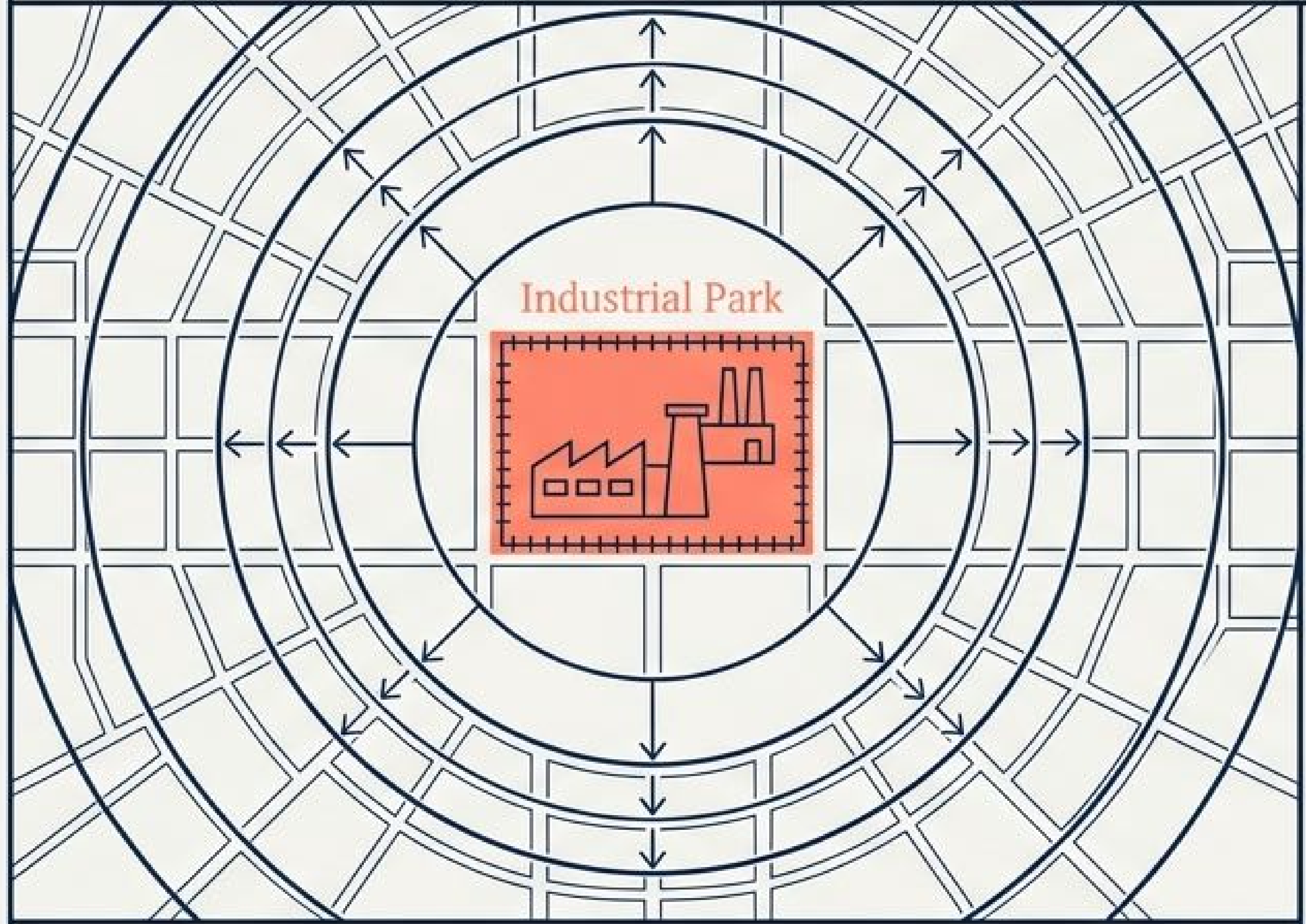
Python implementation covering pyfixest event studies, Callaway-Sant'Anna/Sun-Abraham estimators, Goodman-Bacon decompositions, survey-weighted RCS, and Conley spatial standard errors.

The Methodological Ladder



The Identification Challenge in Evaluating Industrial Policy

The Policy Goal



Fenced zones with serviced land designed to lure factories and lift surrounding economies.

The Endogeneity Problem

Parks are rarely randomly placed.

The Ethiopian government sited them near existing cities and road networks—areas already growing faster.

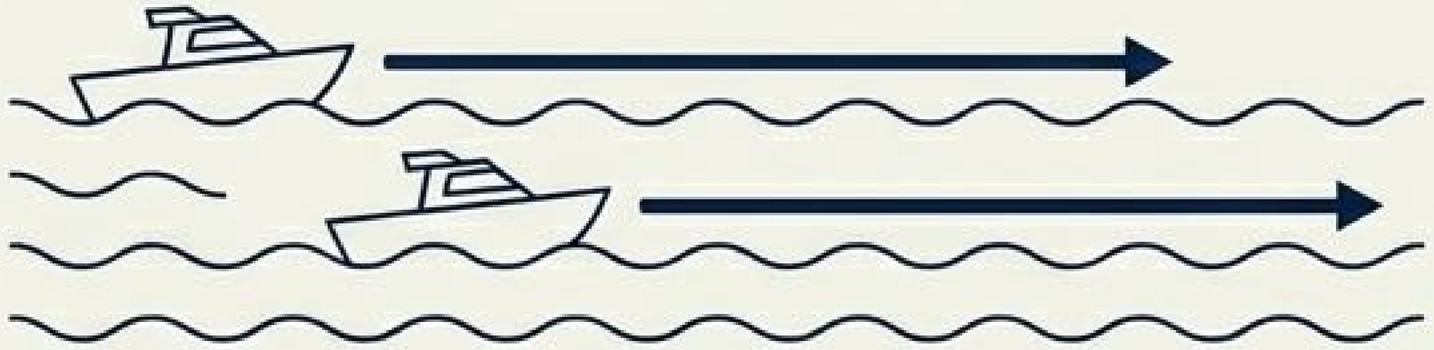
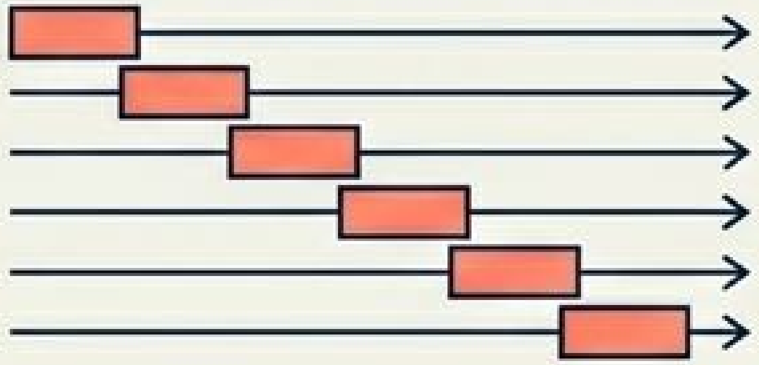
Observational vs. Experimental Data

- **Solution:** A research design that nets out pre-existing differences and handles a staggered 2008-2021 rollout across 17 treated woredas.

Diagnostic Matrix for Aligning Data Architecture to Research Design

	District Layer	**Household Layer**	**Individual Layer**
Data Type	Balanced Panel (Satellite)	Repeated Cross-Section (DHS)	Repeated Cross-Section (DHS)
Sample Size	139 woredas × 16 years (2005–2020)	13,200 households over 5 survey rounds	17,900 individuals
Econometric Constraint	Supports annual event time and genuine panel event studies.	Different respondents per round; no panel key. Requires survey weights and coarse event phases.	Different respondents per round; no panel key. Requires survey weights and coarse event phases.
Permitted Fixed Effects	`Woreda + Region-by-Year`	`District + Region-by-Round (No household FE)`	`District + Region-by-Round`

Core Identification Axioms and Vocabulary

Formal Math	Visual Analogy	Contextual Example
$E[Y_i(1) - Y_i(0) \mid D_i = 1]$	 Measuring the bonus speed on the specific car that received the new engine, not a random car on the street.	The effect (+0.215 light increase) on the 17 woredas that actually got a park.
$E[Y_0(t) - Y_0(\text{pre}) \mid D{=}1] = E[Y_0(t) - Y_0(\text{pre}) \mid D{=}0]$		Confirmed by checking pre-treatment event study leads.
Overlapping 2x2 DiD Matrix	Units adopt treatment at different times, creating a stack of overlapping 2×2 comparisons. 	Ethiopia's parks opened across 8 cohorts (2008 to 2020).

Step 1: Estimating the Static Two-Way Fixed Effects Baseline

Equation Breakdown

$$Y_{dt} = \underbrace{\beta}_{\text{The ATT}} D_{dt} + \underbrace{\alpha_d}_{\text{Woreda Fixed Effect}} + \gamma_{\underbrace{r(d),t}_{\text{Region-by-Year Fixed Effect}}} + \varepsilon_{dt}$$

The ATT

Woreda Fixed Effect
(absorbs permanent district traits, like synthetic bright bases).

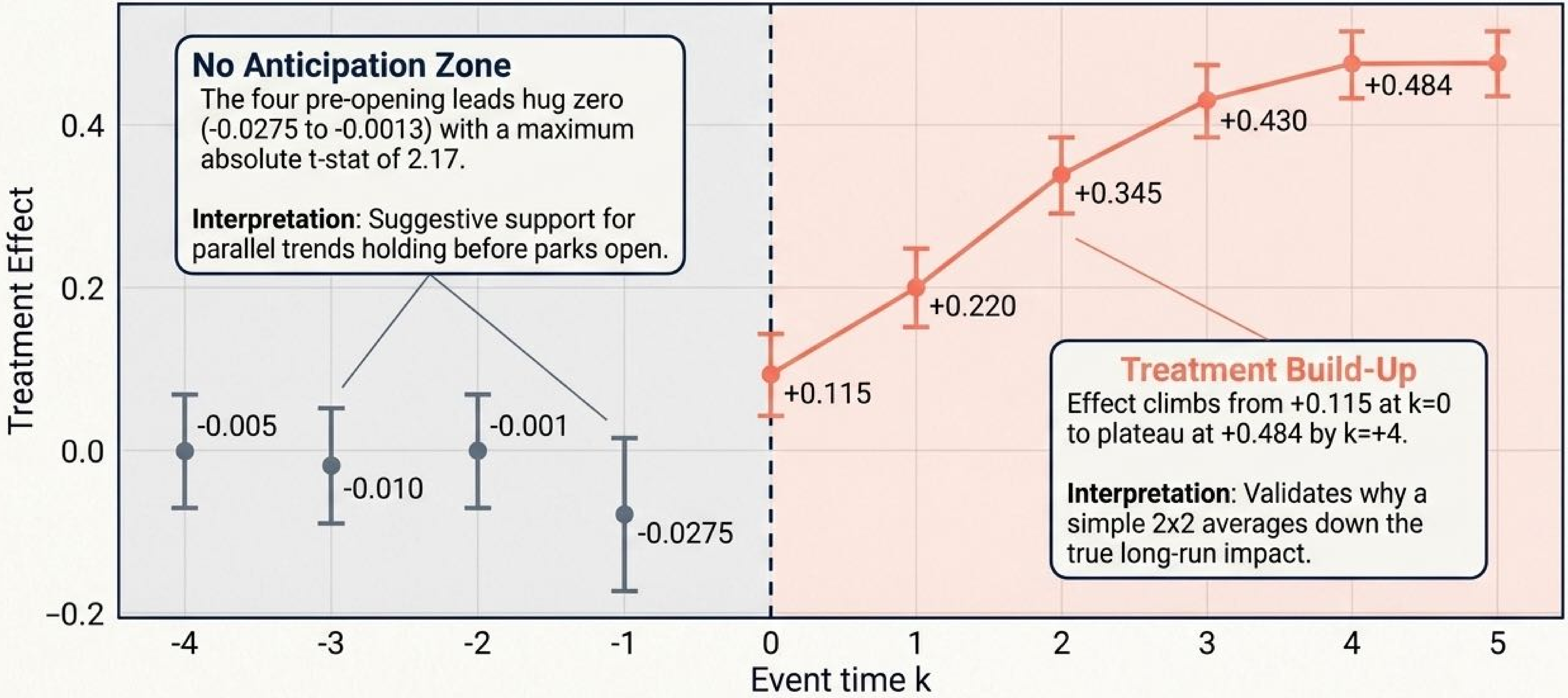
Region-by-Year Fixed Effect
(absorbs shared regional shocks).

```
pyfixest formula:
ihs_light ~ treatment | woreda + region^year
```

A park raises inverse-hyperbolic-sine (IHS) nighttime light by +0.215 ($p < 0.01$).

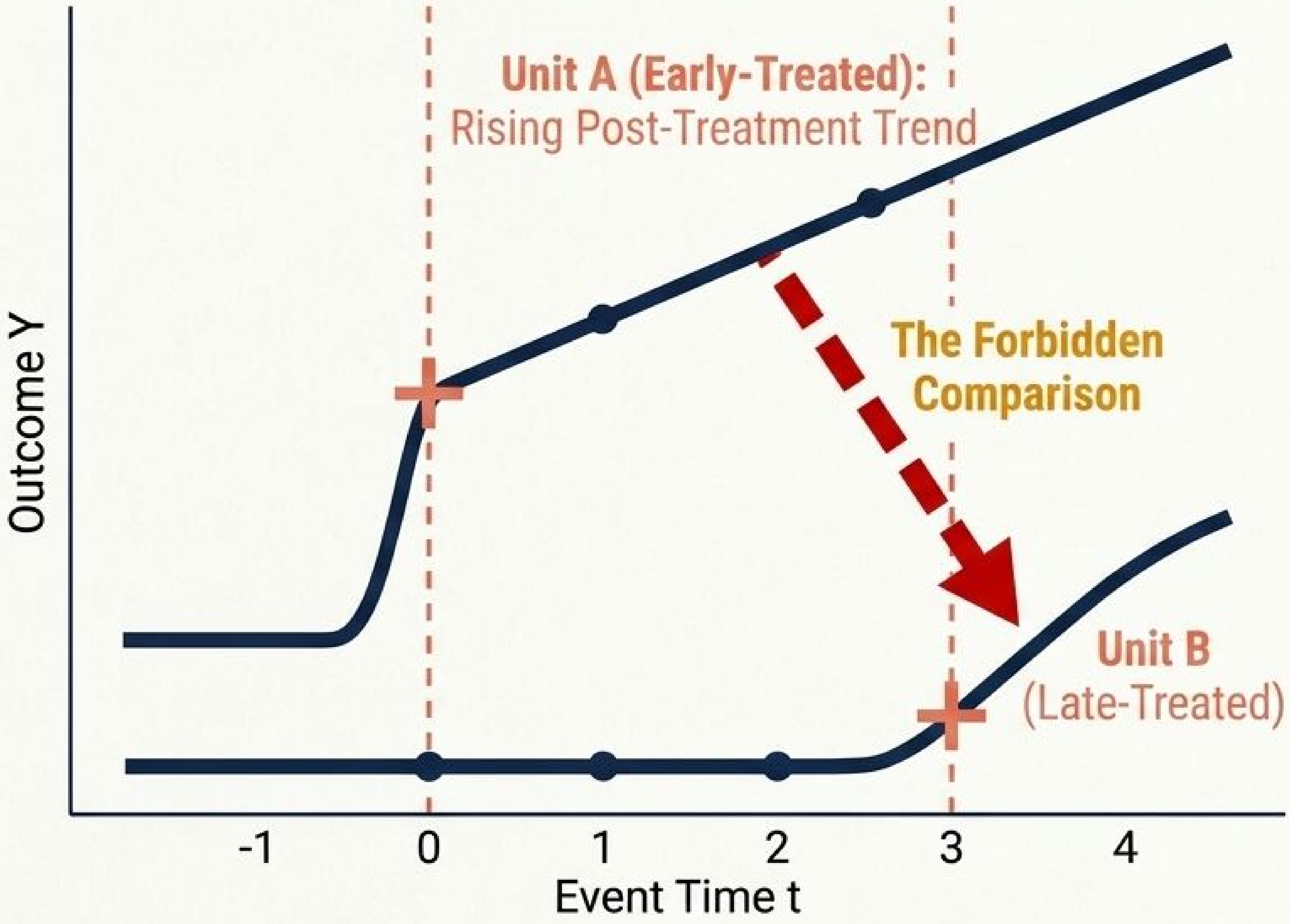
⚠ Note: The naive model without differential pre-trend interactions overstates the effect (+0.270) due to treated districts' faster baseline urbanization.

Step 2: Unpacking Dynamics with the Event Study



NotebookLM

The Staggered Timing Crisis: Forbidden Comparisons



Analogy Callout

Grading on a curve where some students were secretly given the exam early and then used as the "average" everyone else is scored against.

Consequence

When treatment effects grow over time, TWFE **erroneously uses early-treated units' post-treatment trends as the contol as the control baseline for newly-treated units**. These forbidden comparisons receive negative weights and can severely bias, or even sign-flip, the TWFE estimate.

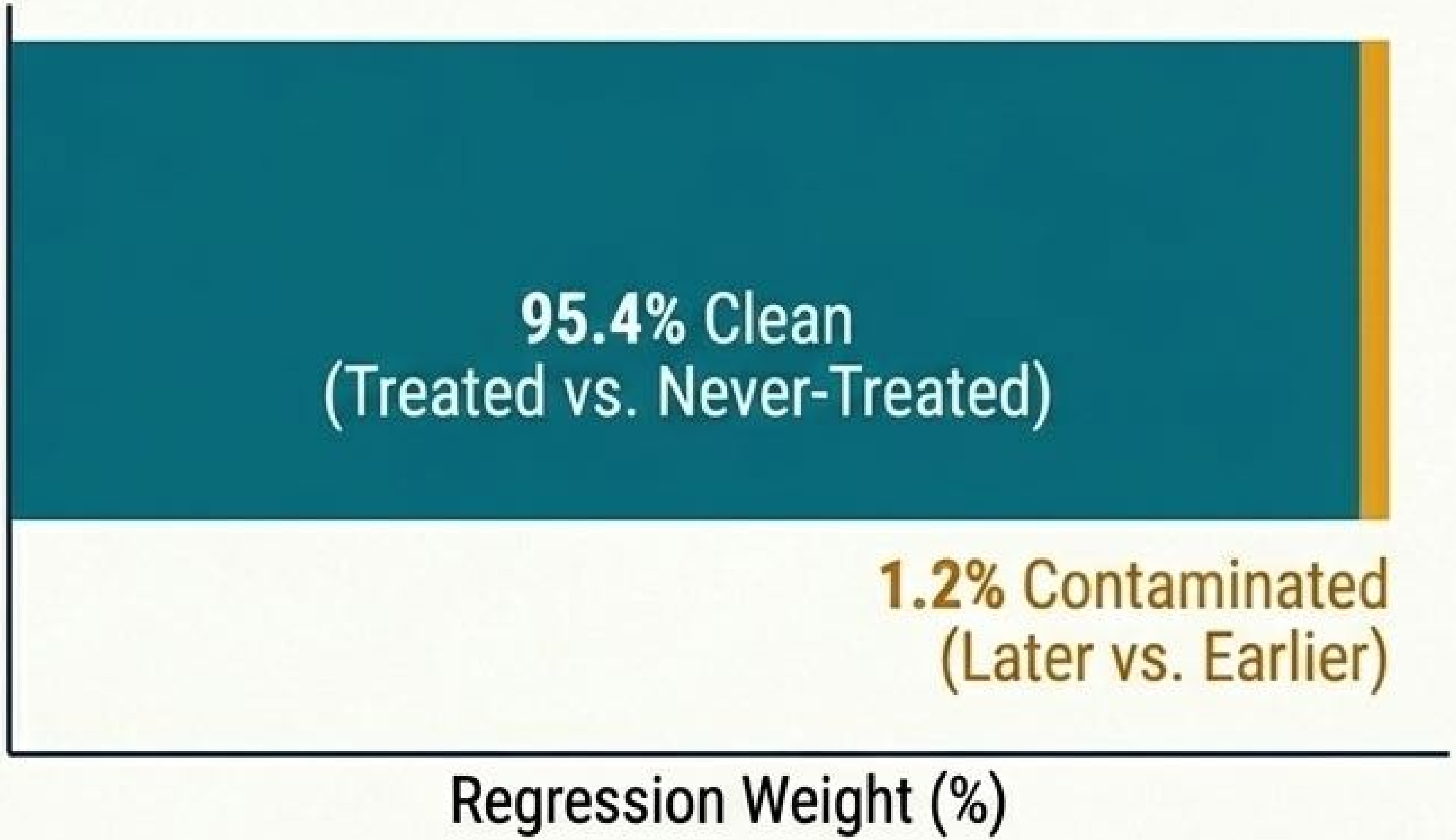
Step 3: Diagnostics & Modern Estimators

The Estimator Matrix

Estimator	Estimate (ATT)
TWFE	+0.270
Callaway-Sant'Anna	+0.256
Sun-Abraham	+0.299
Borusyak/Gardner	+0.302

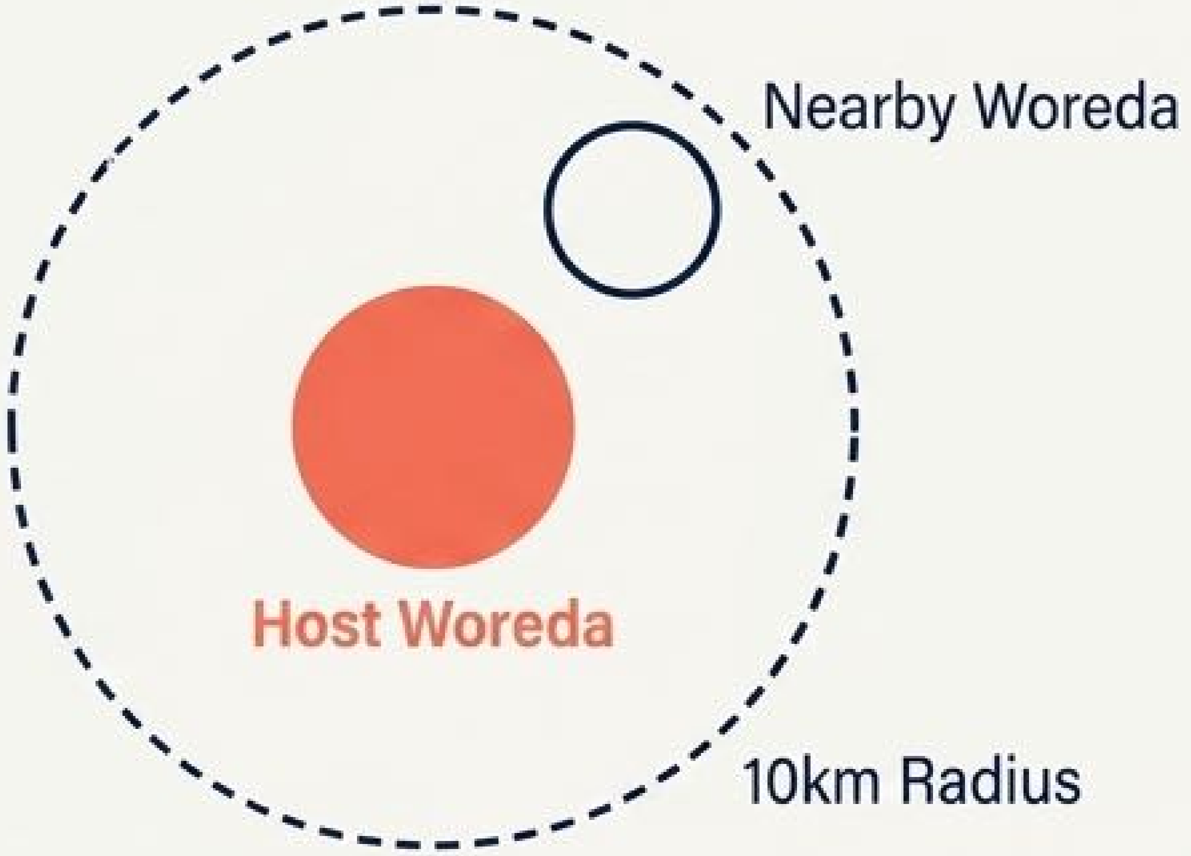
All target the ATT, all are significant at 1%, and they agree within a tight 0.046 spread.

Bacon Decomposition via diff-diff



Conclusion: Plain TWFE is barely contaminated here because a large never-treated pool (122 propensity-score-matched controls) dominates the weighting, explaining why the modern estimators tightly match the baseline.

Step 4: Testing for Geographic Spillovers (SUTVA)



SUTVA (Stable Unit Treatment Value Assumption)

Requires that one unit's treatment does not affect another's outcome. If a park lifts its neighbors, never-treated controls are contaminated.

The Test: Add a "nearby" indicator for control woredas within 10 km of an operational park.

Host Woreda Effect:

+0.2712

(Large, Significant)

Nearby Woreda Effect:

+0.0648 ($t = 1.06$)

(Small, Insignificant)

Interpretation:
No measurable spillover. The park creates net-new activity rather than displacing neighbors, making SUTVA highly plausible.

Step 5: Adapting DiD to Repeated Cross-Sections (RCS)

Constraint: Five DHS rounds (2000-2019) with completely different respondents. No individual tracking possible.

Action 1: Model Setup

Cannot use household fixed effects. Identify the effect off District × Round group means.

Action 2: Fixed Effects & Time Coding

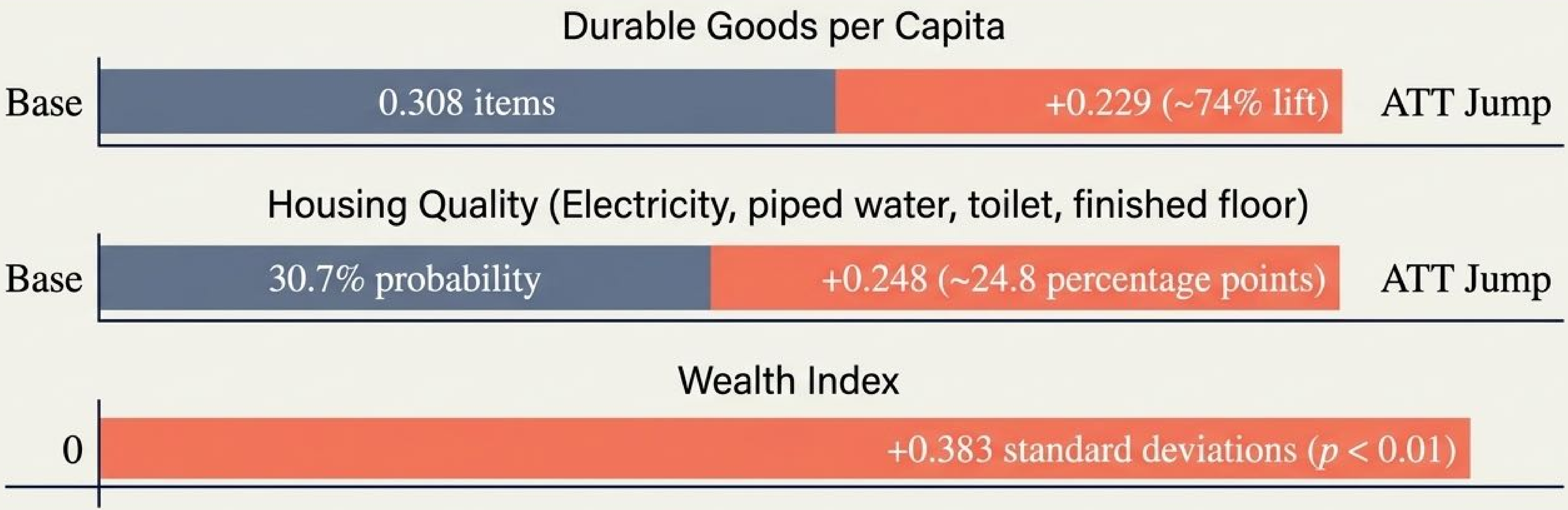
Absorb District + Region-by-Round fixed effects. Replace annual event lags with coarse event phases {-3, ..., +1}.

Action 3: Weighting

Apply complex survey sampling weights.

Core Lesson: RCS DiD compares how neighborhoods shifted relative to each other, even if the individuals walking those neighborhoods changed.

Household Welfare Impacts: Translating Activity to Living Standards



Key Takeaway: Adding household size and head-age controls barely moves the estimates, confirming the district + region-by-round fixed effects effectively absorb primary confounders.

Heterogeneity Analysis: Why Averages Hide the Truth

The Average: Total Non-Agricultural Employment

`+0.091 (ns)`

A naive reading concludes parks failed to create jobs.

Men

`+0.017 (ns)`

Men were largely already off-farm.

Women

`+0.140***`

Parks pull women into factory wage work (textiles/garments).

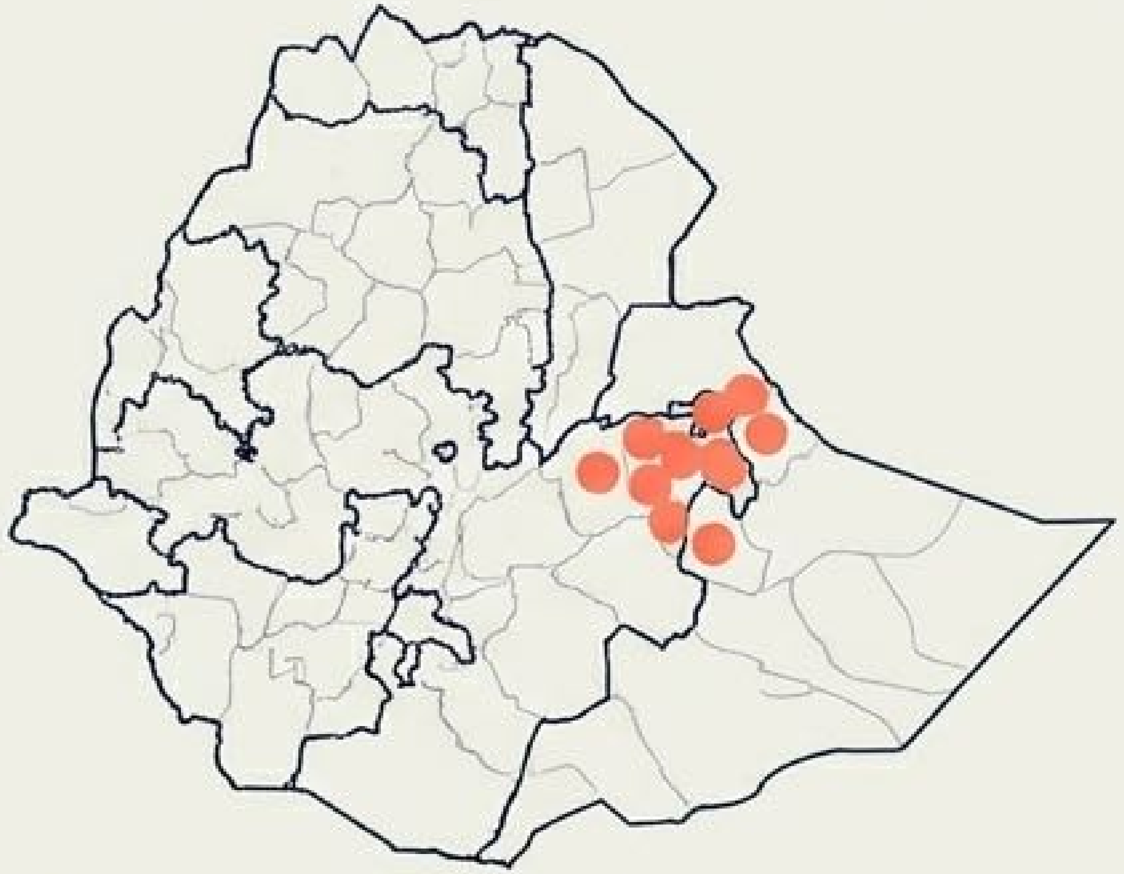
The Empowerment Cascade

Decision-making power: `+0.110`

Savings ownership: `+0.315` (enormous off a 6.3% base)

Acceptance of domestic violence: `-0.210`

Step 6: Advanced Inference with Spatial Standard Errors



The Problem: 17 treated woredas are spatially clustered. Regional shocks hit them simultaneously, violating independence assumptions. Naive standard errors will be too small.

Inference Adjustment

Standard Error Escalation Table	
Naive SE (HC0) :	0.0329
District Cluster SE :	0.0792
Conley Spatial-HAC SE :	**0.0799**

The Solution: Conley spatial-HAC errors allow correlation with itself over time and nearby districts in the same year.

Conclusion: The Conley-HAC SE is **2.43×** larger than the naive SE, yet the **+0.215** ATT remains highly significant (t=**2.69**). Inference is robust.

Synthesis: Policy Formulation and Observational Boundaries

Policy Formulation	Humble Interpretation
• **Site Selection** Effect fades steeply with distance from cities (`-0.0335/km`) and is amplified by paved roads (`+0.6695`). Place-based policy must follow existing connected networks. • **Inclusion Targeting** Place-based success heavily relied on female-intensive sectors. Evaluations must be sex-disaggregated by default.	• **Design Limit** Observational data (non-random placement). Identification rests on parallel trends, which flat pre-trends support but can never mathematically prove. • **Context Limit** The measured ATT applies strictly to these districts with these parameters. • **Data Limit** Analysis relies on synthetically calibrated datasets designed to teach methods.

The Staggered DiD Implementation Checklist

1. **Data Alignment**
Verify panel structure vs. repeated cross-section to determine allowable Fixed Effects.

2. **Baseline Specification**
Run static TWFE to map basic treatment correlations (`pyfixest`).

3. **Dynamic Unpacking**
Plot the Event Study. Look for flat pre-trends ($k<0$) to support no-anticipation.

4. **Staggered Diagnostics** Run Goodman-Bacon decomposition (`diff-diff`).
Deploy SA/CS/BG to rule out negative weighting if forbidden comparisons dominate.

5. **Spillover Checks** Map geographic spillovers to validate the SUTVA assumption.

6. **Robust Inference** Apply spatial-HAC (**Conley**) standard errors if treatment assignment is geographically clustered.